

Ariel Werle

Barendrecht, NL • ariel.werle@gmail.com • arielwrl.github.io • github.com/arielwrl

SUMMARY

Data scientist with a PhD in astrophysics, experienced in delivering end-to-end machine learning solutions. Skilled in statistical modeling, deep learning, and data pipeline development across research and industrial settings.

PROFESSIONAL EXPERIENCE

DATA SCIENTIST September 2024 – Present
Pipesurvey International Zwijndrecht, The Netherlands

- Developed machine learning solutions for the non-destructive detection of gas pipeline defects, guiding maintenance strategies with an 86% success rate.
- Built data analysis tools for process automation, decreasing typical workflow times by 80%.

SCIENTIFIC RESEARCHER March 2020 – August 2024
National Institute of Astrophysics Padova, Italy

- Led the development of a new classification method for hyperspectral imaging of galaxies based on CNNs, reducing false positives to 0.5%.
- Led the application of Bayesian inference models for the determination of physical parameters of star-forming regions.
- Participated in over 35 publications in scientific journals.

TECHNICAL ASSISTANT March 2019 – February 2020
University of São Paulo São Paulo, Brazil

- Developed data processing pipelines for one million objects in the S-PLUS astronomical survey.
- Trained and validated classification models to identify astronomical objects.

EDUCATION

Federal University of Santa Catarina
PhD. in Astrophysics 2015 – 2019
Thesis: Analysis of SDSS Spectra and GALEX Photometry with STARLIGHT: Stellar Populations and Dust in Local Galaxies

- Updated a spectral decomposition code based on MCMC and developed python wrappers.
- Performed numerical modelling and statistical data analysis for a sample of 230 thousand galaxy spectra.

MSc. in Astrophysics 2012 – 2014
Thesis: Post-Starburst Galaxies in the Local Universe

- Classification and statistical analysis in large databases.

BSc. in Physics 2007 – 2011

SKILLS

Programming and Scripting

Python • SQL • Bash • R • MATLAB • HTML • JavaScript

Data Science

Scikit-Learn • TensorFlow • MCMC • XGBoost • Bayesian Inference • Pandas • Spark • SciPy • Plotly • Dash • Seaborn • Streamlit • Databricks • OpenAI

Other

Git • Linux • Docker • Latex • Qt • GitHub Actions

BONUS POINTS

- Languages: English (fluent), Portuguese (native), Italian (fluent), Spanish (basic), currently studying Dutch.
- Courses and workshops: Data science career track (Datacamp), Machine Learning with Python (Coursera), Big data within science and industry (Milano-Bicocca University), Data-driven astronomy (Coursera).
- Vast experience in teaching and tutoring students in math and physics.